VOLTAGE COMPARATORS

- Voltage comparator is to compare the voltage VP with VN and produces either a low voltage VOL or a high voltage VOH

$$v_O = V_{OL} \quad \text{for } v_P < v_N$$
$$v_O = V_{OH} \quad \text{for } v_P > v_N$$

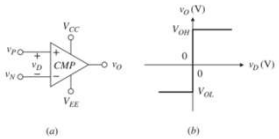


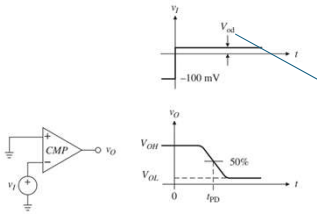
FIGURE 9.1
Voltage-comparator symbolism and ideal VTC. (All node voltages are referenced to ground.)

- With differential input voltage V*D* = *VP* - *VN*
  - *VO* = *VOL* for *VD* < 0 V,
  - *VO* = *VOH* for *VD* > 0 V.
- The voltage transfer curve (VTC), shown in Fig. 9.1*b*, is a nonlinear curve.
- At the origin, the curve is a vertical segment, indicating an infinite gain there, or *VO*/*VD* = ∞.

1

The Response Time

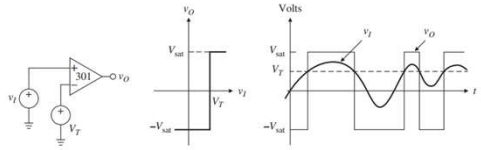
- Comparator speed is characterized in terms of the *response time,* also called the *propagation delay* t_{PD},
  - t_{PD}: Defined as the time it takes for the output to accomplish 50% of its transition in response to a predetermined voltage step at the input.



Though the input step magnitude is typically on the order of 100 mV, its limits are chosen to barely exceed the level required to cause the output to switch states. This excess voltage is called the input overdrive Vod

2

The Op Amp as a Voltage Comparator/Threshold detector



- Opamp to compare *VI* with threshold, *VT* .
  - When *VI* < *VT* ➔ *VO* = -*V*sat ~ = -13 V,
  - When *VI* > *VT* ➔ *VO* = +*V*sat ~ = +13 V.
- Since V*O* goes high whenever V*I* rises above *VT* the circuit is also called a *threshold detector.*
- If *VT* = 0 V, the circuit is referred to as a *zero-crossing detector.*

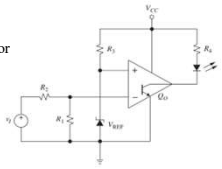
Note: when used as a comparator, the op amp has no control over *VN* due to the absence of feedback. The amplifier now operates in the open-loop mode and, because of its extremely high gain, it spends most of its time in saturation.

3

Level Detectors

- Here we monitor a physical variable that can be expressed in terms of a voltage, and signal
- When it rises above (or drops below) a prescribed value called the *set value,* a specific action is taken.
  - Such as a light-emitting diode (LED) or a buzzer, the turning on of a motor or heater, or the generation of an interrupt to a microprocessor can be performed

Circuit Components
- **Voltage reference** (V_{REF} : A diode (Zener or diode) with bias resistor R_3 (provides a **stable reference voltage** at the inverting input (−).
- **Vtg Divider (R1–R2)**: Scales the input voltage v_I before it reaches the non-inverting input (+).
- **Comparator (Op-amp)**: Compares v_I (scaled) with V_{REF}.
- **LED + resistor** R_4 :Indicates when the comparator switches state.



4

Threshold Derivation

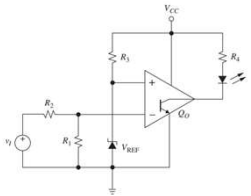
The comparator switches when the two inputs are equal:

$$\frac{R_1}{R_1 + R_2} v_I = V_{REF}.$$

Solving for the threshold input voltage V_T:

$$V_T = \left(1 + \frac{R_2}{R_1}\right) V_{REF}.$$

So, V_T **is the trip point** of the detector.



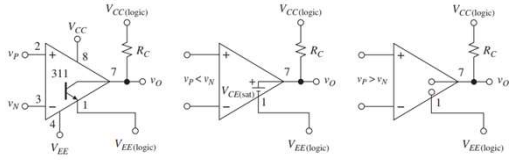
Operation
- **Case 1:** $v_I < V_T$
  - The scaled input (+) is **less than** V_{REF}.
  - Comparator output Q_O goes **OFF** .
  - LED is **OFF** (no current through R_4)
- **Case 2:** $v_I > V_T$
  - The scaled input (+) is **greater than** V_{REF}.
  - Comparator output Q_O goes **ON**
  - Current flows through R_4 and LED → **LED glows**.

Q_O = Off for $v_P > v_N$
Q_O = On for $v_P < v_N$

5

Practical Notes

- R_3 biases the reference diode (provides stable V_{REF})
- R_4 limits the LED current.
- By swapping op-amp inputs (+ and −), the LED would glow when $v_I < V_T$.

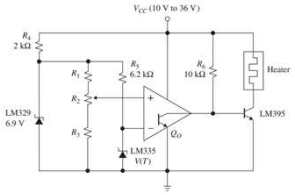


6

On-Off Control

To keep something, like a sensitive electronic component or a small chemical sample, at a constant, specific temperature. It does this by turning a **Heater** ON when the temperature gets too cold and turning it OFF when it gets too hot.

Circuit Overview



•**Sensor** → LM335 (temperature sensor, outputs voltage proportional to absolute temperature).

•**Reference** → LM329 (precision 6.9 V reference diode).

•**Comparator** → LM339 (compares sensor voltage with reference divider voltage).

•**Switching element** → LM395 (protected power transistor, drives the heater).

The goal is to **turn the heater ON when temperature is below setpoint**, and **turn it OFF when above setpoint**.

7

Working Principle

1. Temperature sensor (LM335)

- LM335 outputs a voltage proportional to absolute temperature:

$$V_T = \frac{T}{100} \quad (10\ \text{mV/K})$$

At 300 K (27 °C), $V_T \approx 3.0$ V.

2. Reference source (LM329 + divider)

•The LM329 provides a stable 6.9 V reference.

•Divider R_1, R_2, R_3 scales it down to set the **desired temperature threshold** (setpoint).

3. Comparator (inside LM339)

Compares the sensor voltage (VT) with the reference voltage (VR).

- If $VT < VR$ → output QO is low (transistor off).
- If $VT > VR$ → output saturates high, turning on the heater through LM395.

4. Heater driver (LM395)

- LM395 is a rugged power transistor with built-in protection.
- Acts as a switch to connect the heater to VCC.
- It turns on when the comparator output drives its base.

8

At low temperature

- Sensor voltage VT is below the reference VR
- Comparator output goes high.
- This drives LM395 into saturation (ON).
- Current flows through heater → Heater ON.

At high temperature

- Sensor voltage VT rises above VR.
- Comparator output goes low.
- LM395 turns OFF.
- Heater current is cut → Heater OFF.

9

Window detector

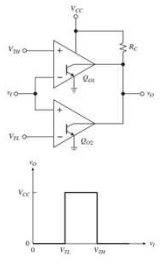
A window detector is designed to determine if an input voltage (v_I) falls within a specific voltage range, or "window," defined by an upper threshold (V_{TH}) and a lower threshold (V_{TL})

•Two thresholds are set:
- V_{TL} →Lower threshold.
- V_{TH} →Upper threshold.

So:

•If $V_{TL} < v_I < V_{TH}$ →Inside the window.

•If $v_I < V_{TL}$ or $v_I > V_{TH}$ →Outside the window.



10

Circuit Operation

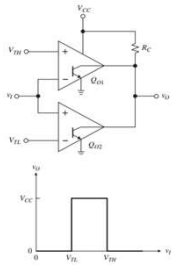
•The circuit uses **two comparators**:
- Q_{O2} :Compares input v_I with V_{TL}.
- Q_{O1} :Compares input v_I with V_{TH}.

•**Case 1: Inside the window** ($V_{TL} < v_I < V_{TH}$)
- Both comparators' transistors are **OFF** .
- R_C pulls v_O up to V_{CC}.
- **So output = High.**

•**Case 2: Input below window** ($v_I < V_{TL}$)
- Comparator Q_{O2} turns ON (because input is less than lower limit).
- Q_{O2} pulls the output down near 0 V.
- **So output = Low.**

•**Case 3: Input above window** ($v_I > V_{TH}$)
- Comparator Q_{O1} turns ON.
- Q_{O1} pulls the output down near 0 V.
- **So output = Low.**

- Therefore, the output is **HIGH only when input lies between the two thresholds.**



VTC

•On the x-axis → Input voltage v_I.

•On the y-axis → Output voltage v_O.

•Output stays HIGH ($\approx V_{CC}$) only between V_{TL} and V_{TH}.

•Outside this window, output drops LOW (≈ 0 V).

11

Feedback in Amplifiers

- Negative feedback → stabilizes, keeps op-amp in the linear region (used in amplifiers, regulators).
- Positive feedback → destabilizes, pushes op-amp into saturation (used in switches, oscillators, Schmitt triggers).

A **Schmitt trigger** is a comparator circuit with **positive feedback**.

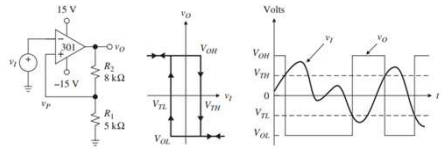
•It produces **hysteresis**: two distinct threshold voltages:
- V_{T+}: upper threshold
- V_{T-} : lower threshold

•Output switches between V_{OH} (logic HIGH) and V_{OL} (logic LOW).

•Once output switches, it stays latched until input crosses the *other* threshold.

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Inverting Schmitt Trigger

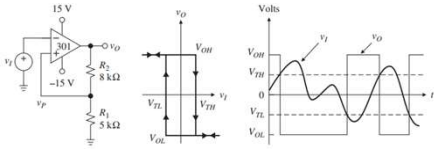


- An Inverting Schmitt Trigger converts a noisy or slowly changing analog input signal into a clean, digital-like output with fast transitions.
- Its key characteristic is **hysteresis**, meaning it has two different switching thresholds:
  - an Upper Threshold (V_{TH}) and
  - a Lower Threshold (V_{TL}).

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Threshold Voltages

From the divider relation:



Upper Threshold (V_{TH})

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_{OH}$$

Lower Threshold (V_{TL})

$$V_{TL} = \frac{R_1}{R_1 + R_2} V_{OL}$$

Hysteresis Width (ΔV_T)

$\Delta V_T = V_{TH} - V_{TL}$

$$\Delta V_T = (V_{OH} - V_{OL}) \frac{R_1}{R_1 + R_2}$$

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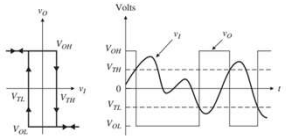
Working

1. **Starting with v_I very low (e.g., negative), so v_O is initially high (V_{OH})**
   - When v_I just exceeds V_{TH} the inverting input becomes slightly higher than the non-inverting input. The op-amp output will rapidly **switch from** V_{OH} **to** V_{OL}

$$V_{OH} \frac{R_1}{R_1 + R_2} = V_{TH}$$

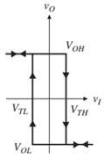
2. **Starting with v_I very high (e.g., positive), so v_O is initially low (V_{OL})**
   - When v_I just falls below V_{TL} ,the inverting input becomes slightly lower than the non-inverting input. The op-amp output will rapidly **switch from** V_{OL} **to** V_{OH}

$$V_{OL} \frac{R_1}{R_1 + R_2} = V_{TL}$$



15

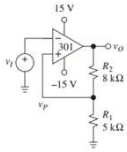
Voltage Transfer Characteristic (VTC)



- When v_I is increasing (moving right on the graph), the output stays at V_{OH} until v_I hits V_{TH} ,then it drops to V_{OL}
- When v_I is decreasing (moving left on the graph), the output stays at V_{OL} until v_I hits V_{TL} ,then it rises to V_{OH}
- Notice that V_{TH} is greater than V_{TL} This "gap" is the hysteresis.

Zero-Crossing Detector:

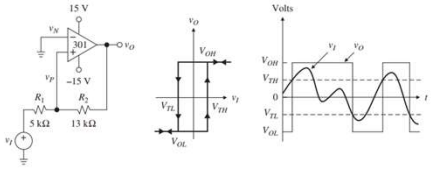
- If $R_1/R_2 \to 0$ meaning R_1 is very small or R_2 is very large, then V_{TH} and V_{TL} would approach 0V. This effectively turns the circuit into an inverting zero-crossing detector, but still with a small amount of hysteresis.



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Noninverting Schmitt Trigger

Noninverting Schmitt Trigger made works very similarly to the inverting type, but the signal is given to the non-inverting input this time, resulting in a different switching action and output polarity.



•The circuit provides two distinct **thresholds**:

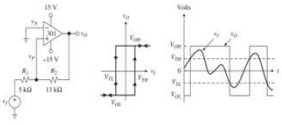
- V_{TH} : Upper threshold: When the input rises high enough, the output jumps up to V_{OH}.
- V_{TL} : Lower threshold: When the input drops low enough, the output snaps down to V_{OL}.

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- For *vI* ≪ 0(- ve), the output will saturate at *VOL*.
- If we want *vO* to switch state, *vI must rise* above V_{TH} .
  - So we can write

$$(V_{TH} - 0)/R_1 = (0 - V_{OL})/R_2$$

$$V_{TH} = -\frac{R_1}{R_2} V_{OL}$$



- Once *vO* has snapped to *VOH*, *vI* must be lowered V_{TL}

$$(V_{OH} - 0)/R_2 = (0 - V_{TL})/R_1$$

$$V_{TL} = -\frac{R_1}{R_2} V_{OH}$$

- The resulting VTC in the *counterclockwise* direction.
- The hysteresis width is

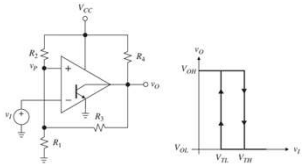
$$\Delta V_T = \frac{R_1}{R_2}(V_{OH} - V_{OL})$$

Note: R1/R2 → 0 we obtain a noninverting zero-crossing detector

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VTC Offsetting

In single-supply Schmitt triggers the need arises to offset the VTC so that it lies entirely within the first quadrant.



•The Schmitt trigger circuit uses an op-amp, resistors R_1, R_2, R_3, R_4 ,and a pull-up network to shape its switching thresholds.
•The output voltage v_o switches sharply between $V_{OH} \approx V_{CC}$ (high state) and $V_{OL} \approx 0$ (low state).
•The input threshold voltages, V_{TL} (lower threshold) and V_{TH} (upper threshold), determine when the output flips state, introducing hysteresis

19

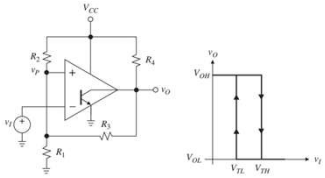
Superposition Principle at at v_p

$$v_P = \frac{R_1 \parallel R_3}{(R_1 \parallel R_3) + R_2} V_{CC} + \frac{R_1 \parallel R_2}{(R_1 \parallel R_2) + R_3} v_O$$

Threshold Calculation
To set the hysteresis:
•For $v_o = V_{OL} \approx 0$,set $v_p = V_{TL}$.
•For $v_o = V_{OH} \approx V_{CC}$,set $v_p = V_{TH}$.

$$V_{TL} = \frac{R_1 \parallel R_3}{(R_1 \parallel R_3) + R_2} V_{CC}$$

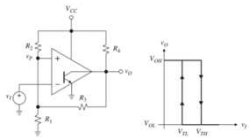
$$V_{TH} = \frac{R_1}{R_1 + (R_2 \parallel R_3)} V_{CC}$$



20

- Simplifying for V_{TH}
- When output $v_o = V_{CC}$,feedback affects the non-inverting input.
- The threshold (V_{TH}) is defined as the input where the output switches back LOW.
- We focus on the voltage divider created by R_1 ,with parallel combination $(R_2 \parallel R_3)$,both connected to V_{CC}:

$$V_{TH} = \frac{R_1}{R_1 + (R_2 \parallel R_3)} V_{CC}$$



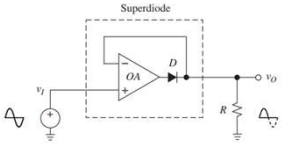
Rearranging V_{TH} and V_{TL}

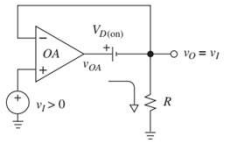
$$\frac{1}{R_2} = \frac{V_{TL}}{V_{CC} - V_{TL}} \left(\frac{1}{R_1} + \frac{1}{R_3}\right) \qquad \frac{1}{R_1} = \frac{V_{CC} - V_{TH}}{V_{TH}} \left(\frac{1}{R_2} + \frac{1}{R_3}\right)$$

Since we have two equations and four unknown resistances, we fix two, say, *R4* and *R3* *R4*, and then solve for the other two.

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PRECISION RECTIFIERS : Half-Wave Rectifiers





Case 1: Positive Input ($v_I > 0$)

- Op-amp output swings positive
- D is forward-biased
- Creating a negative feedback path.
- This enables the virtual short principle, resulting in:

$v_o = v_I$

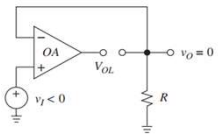
- The op-amp output voltage becomes:
  $V_{OA} = v_o + V_{D(on)} = v_o + 0.7\ V$
- This configuration is called a superdiode because the diode's forward voltage drop is effectively eliminated by the negative feedback.

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Case 2: Negative Input ($v_I < 0$)

- Op-amp output swings negative,
- D is reverse-biased.
- This breaks the feedback loop, causing:

$v_o = 0$



- The op-amp operates in open-loop mode and saturates at:

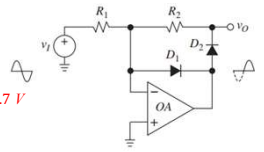
$V_{OA} = V_{OL} \approx -13\ V$ *(with* $V_{EE} = -15\ V$*)*

Limitations

- When transitioning from negative to positive input, the op-amp must recover from saturation and swing from -13 V to v_I + 0.7 V, causing distortion due to slew rate limitations.

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Improved Half-Wave Rectifier



Case 1: Positive Input ($V_I > 0$)
•D_1 conducts, creating negative feedback
•By virtual ground principle: $v^- = 0$
•D_1 clamps the op-amp output at: $V_{OA} = -V_{D1(on)} \approx -0.7\ V$
•D_2 is reverse-biased, so no current flows through R_2
•Output voltage: $v_o = 0$

Case 2: Negative Input ($V_I < 0$)
•Op-amp output swings positive, turning D_2 ON
•D_1 is reverse-biased (off)
•Negative feedback is maintained through D_2 and R_2
•Current balance equation: $\frac{v_o - 0}{R_2} = \frac{0 - v_1}{R_1}$

Solving for the output voltage:

$$v_o = -\frac{R_2}{R_1} v_1$$

The op-amp output voltage becomes:

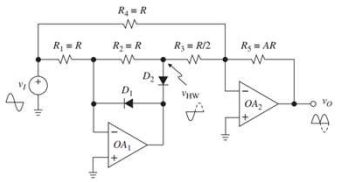
$V_{OA} = v_o + V_{D2(on)}$

The Clever Part:

- When $V_I > 0$, op-amp's output $V_{OA} \approx$ -0.7V.
- It is **not** saturated at -13V.
- So in the other half, we need not over come the saturation voltage of Op-amp as in the previous ckt.
- So this ckt is quicker

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Full-Wave Rectifiers



There are two sections:
1. OA_1 : Half-Wave Rectifier Section
2. OA_2 : Summing Amplifier Section

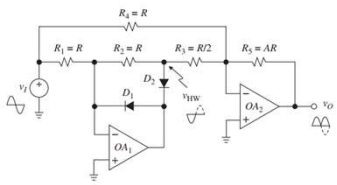
OA_1 (Half-Wave Rectifier Section):
- Provides inverting half-wave rectification
- Output: $v_{HW} = -\frac{R_2}{R_1} v_1$ for $v_I > 0$,
- $v_{HW} = 0$ for $v_I < 0$

OA_2 (Summing Amplifier Section):
- Sums v_I and v_{HW} in specific ratios
- Output equation: $v_o = -\frac{R_5}{R_4} v_1 - \frac{R_5}{R_3} v_{HW}$

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Case 1: Positive Input ($v_I > 0$)

- For positive inputs, v_{HW} = - (R_2/R_1)v_I
- $v_o = A_p v_1$
  where $A_p = \frac{R_2 R_5}{R_1 R_3} - \frac{R_5}{R_4}$



Case 2: Negative Input ($v_I < 0$)

- For negative inputs, $v_{HW} = 0$
- $v_o = A_n v_1$
  where $A_n = \frac{R_5}{R_4}$

Design for Equal Gains
To achieve $A_p = A_n = A$,the circuit uses:
- $R_1 = R_2 = R_4 = R$
- $R_3 = R/2$
- $R_5 = AR$

This results in the absolute value function: $v_o = A \mid v_1 \mid$

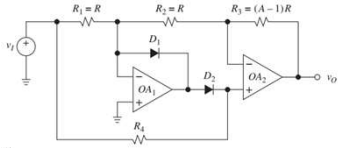
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Tolerance Analysis

- Due to resistor tolerances, the gains A_p and A_n will differ.
- The maximum percentage difference is approximately:
  $$\frac{100 \mid A_p - A_n \mid}{A} \approx 800p$$
  where p is the percentage tolerance.
- For 1% resistors, this can result in up to 8% gain mismatch.
- So we go for alternative FWR realization

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Alternative FWR with Only Two Matched Resistors

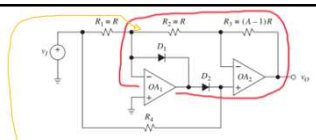


Case 1: Positive Input ($v_I > 0$)
- D_1 conducts, allowing OA_1 to maintain virtual ground at its inverting input
- OA_1 output is ≈ -0.7V
- D_2 is reverse-biased (off)
- R_4 transmits v_I directly to OA_2
- OA_2 acts as a non-inverting amplifier

The gain equation becomes:
$$A_p = 1 + \frac{R_3}{R_2}$$

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Case 2: Negative Input ($v_I < 0$)



- D_1 is reverse-biased (off)
- D_2 is forward-biased
- OA_1 maintains virtual ground through the feedback path D_2-OA_2-R_3-R_2
- The circuit forms a complex feedback network

- Applying Kirchhoff's Current Law (KCL) at the virtual ground node:
  $$\frac{0 - v_1}{R_1} = \frac{v_o - 0}{R_2 + R_3}$$
- Solving for v_o
  $$v_o = -A_n v_1$$
  where $A_n = \frac{R_2 + R_3}{R_1}$

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Design for Equal Gains

Setting $A_p = A_n = A$ requires:
- $R_1 = R_2 = R$
- $R_3 = (A-1)R$

This yields:
$$A_p = 1 + \frac{(A-1)R}{R} = A$$
$$A_n = \frac{R + (A-1)R}{R} = A$$

The final transfer function becomes: $v_o = A \mid v_1 \mid$

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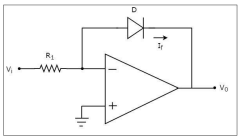
Log Amplifier using Transistor

A **logarithmic amplifier**, or **log amplifier**, is an electronic circuit designed to produce an output voltage that is **proportional to the logarithm of the applied input voltage**

The **nodal equation** at the inverting input terminals node is

$$\frac{0 - V_i}{R_1} + I_f = 0$$

$$\Rightarrow I_f = \frac{V_i}{R_1} \quad (1)$$



The **equation for current** flowing through a diode, when it is in forward bias

$$I_f = I_s e^{\left(\frac{V_f}{nV_T}\right)} \quad (2)$$

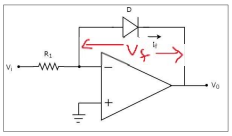
Is: is the saturation current of the diode,
Vf : is the voltage drop across diode, when it is in forward bias,
VT: is the diodes thermal equivalent voltage.

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The **KVL equation** around the feedback loop of the op amp will be

$$0 - V_f - V_0 = 0$$

$$\Rightarrow V_f = -V_0$$



Substituting the value of Vf in eq. 2

$$I_f = I_s e^{\left(\frac{-V_0}{nV_T}\right)} \quad (3)$$

Observe that the left hand side terms of both equation 1 and equation 3 are same.

$$\frac{V_i}{R_1} = I_s e^{\left(\frac{-V_0}{nV_T}\right)}$$

$$\frac{V_i}{R_1 I_s} = e^{\left(\frac{-V_0}{nV_T}\right)}$$

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Applying **natural logarithm** on both sides

$$In\left(\frac{V_i}{R_1 I_s}\right) = \frac{-V_0}{nV_T}$$

$$V_0 = -nV_T In\left(\frac{V_i}{R_1 I_s}\right)$$

The parameters **n, VT** and **Is** are constants. So, the output voltage **V0** will be proportional to the natural logarithm of the input voltage **Vi** for a fixed value of resistance **R1**

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Anti-Logarithmic Amplifier

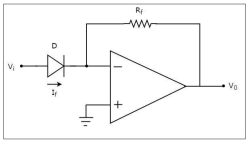
An **anti-logarithmic amplifier**, or an **anti-log amplifier**, is an electronic circuit that produces an output that is proportional to the anti-logarithm of the applied input.

The **nodal equation** at the inverting input terminals node is

$$-I_f + \frac{0 - V_0}{R_f} = 0$$

$$\Rightarrow -\frac{V_0}{R_f} = I_f$$

$$\Rightarrow V_0 = -R_f I_f \quad (1)$$



We know that the equation for the current flowing through a diode, when it is in forward bias

$$I_f = I_s e^{\left(\frac{V_f}{nV_T}\right)}$$

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I_f in eq. 4 $\quad V_0 = -R_f \left\{ I_s e^{\left(\frac{V_f}{nV_T}\right)} \right\}$

$$V_0 = -R_f I_s e^{\left(\frac{V_f}{nV_T}\right)} \quad (5)$$

The KVL equation at the input side of the inverting terminal of the op amp will be

$$V_f = V_i$$

Substituting, the value of in the Equation 5, we get

$$V_0 = -R_f I_s e^{\left(\frac{V_i}{nV_T}\right)}$$

Note that, in the above equation the parameters **n, VT** and **Is** are constants. So, the output voltage **V0** will be proportional to the anti-natural logarithm (exponential) of the input voltage **Vi** , for a fixed value of feedback resistance **Rf**

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